

Operating and Maintenance Manual
and Building Log Book for the
Mechanical Services Installation at
**General Office Building
Reception Refurbishment**

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Volume One of One

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SECTION ONE

GENERAL DESCRIPTION OF INSTALLATION

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Title: Introduction	Item: 1.1.1
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The purpose of this operating and maintenance manual and building log book is to offer the personnel concerned with the operation and maintenance of the services within the building, assistance in the performance of their duties. It is intended that this manual will be used to assist skilled engineers, experienced in the operation and maintenance of building services, to provide them with a comprehensive working knowledge of the various systems installed within the project.

Although the manual functions as a complete entity, each section is self contained and provides information on differing aspects of the installation. Familiarisation with all sections prior to formulating an operating and maintenance regime will prepare the building maintenance personnel for the occasions when a particular problem occurs.

The information, procedures and instructions contained herein are those appertaining at the time of preparation and the date of issue. Changes arising in building usage, legislation, etc, will necessitate revisions to be made in order to retain the manual's validity and usefulness. Any alterations to this document (addition, revision or omission) should be recorded on the following page.

The information within this manual will allow maintenance to be carried out safely and reliably, as well as providing important data upon which calculations may be made for design of alterations and/or additions, should the need arise. It should be noted that this manual is not intended to supersede or conflict with any standard maintenance/inspection routines already in use.

This manual has been based on the content suggested by the Building Services Research and Information Association in their application guide AG 1/87.1 (Operating and Maintenance Manuals for Building Services Installations). Where issued electronically the same organisation's guide BG 2/2004 (Computer Based Operating and Maintenance Manuals) has been utilised (their suggested "PDF Based System"). This manual has also been compiled to cater for other industry legislation requirements such as Health and Safety Executive (CDM) and the Building Regulations (Part L2) *et al.*

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This manual is also available as a fully interactive electronic document suitable for direct use on a computer
For further details or to obtain a quotation, please contact the author with your requirements quoting the above reference number.

SECTION 01 E**General Description of Installation**

Title: Document Revisions	Item: 1.1.2
Category: Preliminary Items	Page: 1 of 1

Initial Production	
Stage of Development (all dates refer to author)	Date
Draft Copy Production Commenced	21/5/2007
Draft Copy Issued	23/5/2007
Comments Received, Information/Clarification Sought	25/6/2007
Information/Clarification Received, Final Copy Production Commenced	27/6/2007
Final Copy Issued	28/6/2007

Subsequent Revision History	
Date	Description of Revisions

Title: Purpose of Each Section	Item: 1.1.3
Category: Preliminary Items	Page: 1 of 1

Section One - General Description of Installation - This section contains a general description of each system, together with other introductory information. The descriptions are intended to be read in conjunction with the Record Drawings to aid the maintenance personnel in familiarising themselves within the installation, prior to formulating an operating and maintenance programme.

Section Two - Plant Data Schedules - This section contains the technical data for each item of equipment extracted from the manufacturers' literature and other sources to assist with the maintenance and possible replacement of faulty items of equipment. Recommended stocked spares list, lubricants, special tools and equipment are also included.

Section Three - Emergency Procedures and First Aid - This section contains the procedures that should be followed in the event of an emergency, together with the emergency call-out telephone numbers for the local utility companies and basic first aid instructions.

Section Four - Health and Safety Information - This section contains check lists and safety procedures that should be followed to ensure that the operation and maintenance of the installation is undertaken in a safe manner.

Section Five - Operating Procedures - This section contains all necessary operational procedures for the systems. It is intended that the procedures are read in conjunction with the manufacturers' literature so that operational procedures are extended beyond each item of equipment and cover the full system.

Section Six - Maintenance Schedules - This section contains schedules of maintenance tasks for each item of equipment. It is intended for use in the formulation of a Planned Preventative Maintenance regime. The schedules are intended to be read in conjunction with the manufacturers' literature for detailed maintenance instructions.

Section Seven - Maintenance Instructions and Log Books - This section contains various instructions and procedures necessary when implementing an operating and maintenance regime, including risk assessments, handling/disposal instructions, decontamination procedures and recommendations on the use and keeping of energy usage and maintenance log books, together with sample log book documentation.

Section Eight - Fault Tracing Charts - This section contains charts to assist with the location of faults that occur with systems. Once the appropriate faulty item of equipment has been located, the fault tracing chart within the manufacturers' literature should be consulted to remedy the problem.

Section Nine - Manufacturers' Literature - This section contains the manufacturers' literature for each item of equipment. This information should be used in conjunction with various sections as described above.

Section Ten - Test Certificates and Commissioning Data - This section contains the test certificates and commissioning data that should be consulted when carrying out maintenance tasks, or fault tracing procedures, to ensure the correct and efficient operation of each system.

Section Eleven - Specialist Services Documentation - This section contains the small independent manuals from the specialist sub-contractors which contain all necessary operating and maintenance information.

Section Twelve - Record Drawings - This section contains a copy of each of the Record Drawings which should be read in conjunction with the general descriptions and are intended to assist the maintenance personnel with the familiarisation of the installation.

Title: Schedule of Abbreviations	Item: 1.1.4
Category: Preliminary Items	Page: 1 of 2

Type	Abbreviation	Meaning	Abbreviation	Meaning
Equipment	AHU	Air Handling Unit	EF	Extract Fan
	FCU	Fan Coil Unit	VAV	Variable Air Volume
	SD	Smoke Damper	FD	Fire Damper
	MD	Motorised Damper	VCD	Volume Control Damper
	SG	Supply Grille	SD	Supply Diffuser
	LSD	Linear Slot Diffuser	EG	Extract Grille
	TG	Transfer Grille	WL	Weather Louvre
	RC	Roof Cowl	IV	Isolating Valve
	DRV	Double Regulating Valve	CS	Commissioning Set
	NRV	Non-Return Valve	STR	Strainer
	DC	Drain Cock	DX	Direct Expansion
	GC	Gas Cock	SC	Stop Cock
	DCV	Double Check Valve	PRV	Pressure Relief Valve or Pressure Reducing Valve
	SV	Safety Valve	BV	Ball Valve

Title: Schedule of Abbreviations	Item: 1.1.4
Category: Preliminary Items	Page: 2 of 2

Type	Abbreviation	Meaning
Systems	LTHW	Low Temperature Hot Water
	MTHW	Medium Temperature Hot Water
	CHW	Chilled Water
	VRV	Variable Refrigerant Volume
	MWS	Mains Water Service
	CWDS	Cold Water Down Service
	HWS	Hot Water Service
	BMS	Building Management System
	F&R	Flow and Return
	CF	Cold Feed
	OV	Open Vent
	F&E	Feed and Expansion

Type	Abbreviation	Meaning	Abbreviation	Meaning
Units	°C	Degree Celsius	db	dry bulb
	wb	wet bulb	% RH	Relative Humidity
	W	Watts	kW	kilo Watts
	mm	milli-metres	m	metres
	kg	kilo grammes	Pa	Pascals
	kPa	kilo Pascals	m/s	metres per second
	m³/s	cubic metres per second	kg/s	kilo grammes per second
	kg/h	kilo grammes per hour	l/s	litres per second
	rpm	revolutions per minute	Hz	Hertz
	V	Volts, Voltage	A	Amps

Title: Schedule of Services Identification	Item: 1.1.5
Category: Preliminary Items	Page: 1 of 1

The services installed on this project have been identified by coloured bands in accordance with BS 1710 (1984). The following table illustrates this colour code:-

Service	Basic Colour	Coloured Strips
PIPEWORK		
Mains Water	Green	Auxiliary Blue
Natural Gas	Yellow Ochre	Yellow
Steam	Silver Grey	
Low Temperature Hot Water	Green	Blue/Crimson/Blue
Medium Temperature Hot Water	Green	Crimson/Blue/Crimson
Chilled Water	Green	White/Emerald Green/White
Refrigerant R22	Yellow Ochre	Sea-Green
Refrigerants R134a/R407c/R410a	Yellow Ochre	Emerald Green
Cold Water Down Service	Green	White/Blue/White
Hot Water Service	Green	White/Crimson/White
Drainage	Black	
Compressed Air	Light Blue	
DUCTWORK		
Fresh Air	Green Triangle	
Conditioned Air	Red/Blue Triangle	
Warm Air	Yellow Triangle	
Hot Supply Air	Red Triangle	
Cold Supply Air	Blue Triangle	
Exhaust Air	Grey Triangle	
Foul Air	Brown Triangle	
Contaminated/Hazardous Air	Yellow/Black Triangle	

Title: Specific Health and Safety Issues	Item: 1.1.6
Category: Preliminary Items	Page: 1 of 1

The following information is a summary of the key Health and Safety issues regarding the project covered by this document. It is imperative that the main Health and Safety section later within this document and the main project Health and Safety File are fully consulted.

Residual Risks

The project has been designed to avoid there being any residual risks.

Special Access Requirements

The project has been designed to avoid the need for any special access requirements.

Special Maintenance Considerations

The project has been designed to avoid the need for any special maintenance considerations.

Title: Incoming Services Location	Item: 1.1.7
Category: Preliminary Items	Page: 1 of 1

The main incoming services are located in the following locations:-

- Gas - Not applicable to the contract covered by this manual
- Water - Not applicable to the contract covered by this manual
- Electricity - Refer to electrical manual.

Title: Emergency Telephone Numbers	Item: 1.1.8
Category: Preliminary Items	Page: 1 of 1

The following telephone numbers are for emergency use only:-

Gas	-	Transco	-	0800 111999
Water and Sewage	-	Thames Water	-	0800 714614
Electricity	-	EDF Energy (London)	-	0800 028 0247
Police				
Fire Brigade	-	Emergency Services Operator	-	999
Ambulance				

SECTION 01 E**General Description of Installation**

Title: Schedule of Project Team Members	Item: 1.1.9
Category: Preliminary Items	Page: 1 of 2

Project Team Role	Company
Client	A N Other Client Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678
Employers Agent	A N Other Employers Agent Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678
Architect, Project Manager and Contract Administrator	A N Other Architect Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678
Planning Supervisor	A N Other Planning Supervisor Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678
Property Consultant	A N Other Consultant Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678

Title: Schedule of Project Team Members	Item: 1.1.9
Category: Preliminary Items	Page: 2 of 2

Project Team Role	Company
Consultant Engineer	A N Other Consultant Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678
Main Contractor	A N Other Builder Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678
Services Contractor	A N Other Contractor Ltd 123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678

Title: Brief Description of Installation	Item: 1.1.10
Category: Preliminary Items	Page: 1 of 1

The following is an outline description of the services installed under the contract covered by this document; for a full description of each system reference should be made to the following pages.

Heating and Cooling Systems

- a) A direct expansion heat pump split air conditioning system consisting of a void mounted ducted internal unit which is connected to an air cooled condensing unit located on the roof.

Specialist Services

The contract covered by this document also encompassed various specialist services/systems as detailed below, which were supplied, installed, tested and commissioned by the specialist sub-contractor indicated. For full information relating to each system reference should be made to the self-contained specialist manuals located within Section Eleven.

Mechanical Services Control System - A N Other Controls Company Ltd

Title: Design Criteria	Item: 1.1.11
Category: Preliminary Items	Page: 1 of 1

The following criteria were used during the design of the installation:-

External Conditions

Summer Temperature	-	31.0°C db, 21.0°C wb
Winter Temperature	-	-4.0°C saturated

Internal Conditions

Summer Temperature	-	22.0°C +/- 2.0°C, No Humidity Control
Winter Temperature	-	21.0°C +/- 2.0°C, No Humidity Control

Title: Air Conditioning System	Item: 12.1
Category: Heating and Cooling Systems	Page: 1 of 1

A direct expansion heat pump split air conditioning system has been installed to provide both heating and cooling to the Reception. The internal unit is void mounted above the reception desk and incorporates a washable panel filter, a direct expansion heating/cooling coil and a direct driven centrifugal fan unit. Air is drawn into the unit from the false ceiling void via a shadow gap in the ceiling, heated or cooled as required and supplied back to the space via ranges of galvanised steel spiral ductwork which serve plenum boxes located above a shadow gap on the opposite side of the space.

Liquid and suction refrigeration pipework extends from the internal unit to run through the false ceiling void of the ground floor on a series of galvanised steel cable tray. The pipework then enters a building services riser shaft where it rises through the building onto the roof and connects onto an air cooled condensing unit. The condensing unit incorporates a direct driven propeller condenser fan and a refrigerant compressor.

The internal unit is controlled under the dictates of a return air sensor located within the return air path. Condensate from the internal unit discharges into a gravity condensate drain run within the false ceiling void which, in turn discharges into the existing above ground drainage system.

The pipework is insulated with a closed cellular foam insulation in order to reduce heat gain to the system and vapour sealed to prevent the formation of condensation on the pipe surface.

For the locations of the equipment and the routes of the pipework reference should be made to the Record Drawings contained within Section Twelve.

SECTION 11

General Description of Installation

Title: Specialist Services	Item: 1.3.1
Category: Miscellaneous Systems	Page: 1 of 1

The contract covered by this document also encompassed various specialist services/systems as detailed below, which were supplied, installed, tested and commissioned by the specialist sub-contractor indicated. For full information relating to each system reference should be made to the self-contained specialist manuals located within Section Eleven.

Mechanical Services Control System - A N Other Controls Company Ltd

SECTION TWO

PLAY T DATA SCHEDULES

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<u>ITEM</u>		<u>DESCRIPTION</u>
2.1	-	<u>MISCELLANEOUS ITEMS</u>
2.1.1	-	Directory of Manufacturers
2.1.2	-	Spare Parts Policy
2.1.3	-	Recommended Stocked Spares
2.1.4	-	Lubricants
2.1.5	-	Special Tools and Equipment
2.2	-	<u>HEATING AND COOLING SYSTEMS</u>
2.2.1	-	Air Conditioning Units

SECTION TWO

Title: Directory of Manufacturers	Item: 2.1.1
Category: Miscellaneous Items	Page: 1 of 1

Name	Address	Equipment Supplied
A N Other Controls Company Ltd	123 High Road London E1 2AB Tel: 020 7123 1234 Fax: 020 7123 5678	Mechanical Services Control System
Mitsubishi Electric Europe Ltd	Orbital One Unit 8 & 9 Green Street Green Road Dartford Kent DA1 1QG Tel: 01322 628080 Fax: 01322 628081	Air Conditioning Units

Title: Spare Parts Policy	Item: 2.1.2
Category: Miscellaneous Items	Page: 1 of 1

A majority of the manufacturers' literature detail a full parts list, commonly with exploded view diagrams, however, these lists are not intended to be interpreted as spares which should be stocked, rather a list of the part numbers should a spare need to be ordered.

The determination of what spares at what quantities should be stocked will depend upon a number of factors, including:-

- i) The degree of in house or sub-contracted maintenance.
- ii) The economic factors.
- iii) The implications of system failure to the occupiers of the premises.
- iv) The level of back-up or redundancy offered by the systems.
- v) The lead time of spares.

In order to determine the list of spares which should be stocked the following questions should be considered.

- 1) What parts of the system are consumable and will require routine replacement for the safe and efficient operation of the system. Such items include filters, drive belts, fuses and lamps.
- 2) What systems are critical to the operation of the business, consideration should be given to the stocking of spare motors for such systems. Particularly critical systems will usually have a certain level of redundancy designed into the system, commonly in the form of run and standby fans/pumps, but occasionally with 100% system backup. In such situations the stocking of spare motors may not be necessary.
- 3) What systems are less critical but which involve extended lead times for delivery of spares.

Once a list of which items need to be stocked has been established, consideration needs to be given to the quantity of stock to be held. Important points include:-

- a) The shelf life of spares, for instance drive belts generally have an expiry date.
- b) How often will it be necessary to order new items, the disposable panel filters in ventilation systems may need to be changed every two to three months.
- c) How much storage space is available.
- d) Is secure space available for the storage of high value items.

The following section details the minimum level of spares that are recommended to be stocked, to this should then be added the additional items determined as a result of the consideration of the above factors.

SECTION TWO

Title: Recommended Stocked Spares	Item: 2.1.3
Category: Miscellaneous Items	Page: 1 of 1

The equipment installed on this project do not incorporate any of the following items:-

- i) Consumable parts
- ii) Parts which break easily
- iii) Parts which involve long delivery periods

therefore, there is no requirement for the stocking of spares on such grounds.

SECTION TWO

Plant Data Schedules

Title: Lubricants	Item: 2.1.4
Category: Miscellaneous Items	Page: 1 of 1

There is no plant and equipment on this project which require lubrication.

Title: Special Tools and Equipment	Item: 2.1.5
Category: Miscellaneous Items	Page: 1 of 1

Beyond those tools found within a “standard” tool box very few special tools or equipment are necessary to carry out the operation and maintenance of the systems covered by this manual. Such additional items are as follows:-

Special Tools

- No special tools.

Equipment

- Inclined manometer
- Vane anemometer
- Sling psychrometer
- RPM strobe light
- Electrical multi-meter

SECTION TWO

Title: Air Conditioning Units	Item: 2.2.1
Category: Heating and Cooling Systems	Page: 1 of 2

Internal Unit

General	-	Manufacturer	-	Mitsubishi Electric Europe
	-	Model Number	-	PEAD-RP71EA
	-	Type	-	Ceiling Concealed
	-	Duty	-	6.4 kW
		-	Cooling	
		-	Heating	6.8 kW
	-	Air Volume	-	25.0 m ³ /min
	-	Dimensions	-	1070 mm
		-	Length	
		-	Width	740 mm
		-	Height	307 mm
	-	Weight	-	44.0 kg
	-	Refrigerant Type	-	R410a
	-	Electrical Supply	-	240 volts 1 phase 50 Hz

SECTION TWO

Title: Air Conditioning Units	Item: 2.2.1
Category: Heating and Cooling Systems	Page: 2 of 2

External Unit

General	-	Manufacturer	-	Mitsubishi Electric Europe
	-	Model Number	-	PUHZ-RP71VHA
	-	Dimensions	-	950 mm
		-	Length	
		-	Width	360 mm
		-	Height	943 mm
	-	Weight	-	75.0 kg
	-	Refrigerant Type	-	R410a
		-	Type	
		-	Charge	3.5 kg
	-	Starting Current	-	5.0 amps
	-	Full Load Current	-	10.17 amps
	-	Electrical Supply	-	240 volts 1 phase 50 Hz

SECTION THREE

EMERGENCY PROCEDURES AND FIRST AID

INDEX

<u>ITEM</u>	<u>DESCRIPTION</u>
3.1	- <u>EMERGENCY PROCEDURES</u>
3.1.1	- Introduction
3.1.2	- Failure of Electrical Supply
3.1.3	- Refrigerant Leaks
3.1.4	- Fire
3.2	- <u>FIRST AID PROCEDURES</u>
3.2.1	- Principles and Practice
3.2.2	- Life Saving Measures
3.2.3	- Emergency Situations
3.2.4	- Dressings
3.2.5	- Treatment for Electric Shock
3.2.6	- Further Reading

SECTION THREE

Title: Introduction	Item: 3.1.1
Category: Emergency Procedures	Page: 1 of 1

The emergency procedures are intended as a guide only, each emergency situation that arises has to be taken in the light of its degree of severity and the conditions prevailing at the time.

When an emergency situation is encountered, it is imperative that calm prevails. Measures and actions taken when made in haste or panic, will not only in all probability be wrong, but could spread panic in other people.

A few minutes spent in rationalising the situation and planning your campaign of actions, even in an emergency of the utmost gravity, will save time and possibly injury in the long run.

Never take risks in an emergency, for a risk can turn a minor emergency into a major one. The safety of personnel must always come first.

Any temporary repairs effected or arrangements made during or after an emergency must be corrected as soon as it is practically possible.

SECTION THREE

Title: Failure of Electrical Supply	Item: 3.1.2
Category: Emergency Procedures	Page: 1 of 1

Description: This describes the general steps to take on failure of the electrical supply.

Notes:

Step	Action	Notes
1	Ascertain whether the failure is local or general.	Check whether general lighting and power is affected.
2	If failure is local, check whether fuses have blown, or circuit breaker has tripped.	Replace with correct size fuse, or reset circuit breaker as required, after checking cause of failure and retesting the circuit.
3	If failure is general, check position of lifts (if applicable) and hand winch to nearest lower floor.	Isolate lift supply prior to winching. Inform local electricity company of failure. Place "Lift Out of Service" notices on all floors.
4	Repairs to electrical switchgear and cables should only be undertaken by a fully qualified electrician.	
5	If the failure is general, isolate the supply to items of plant until the supply is restored.	

Title: Refrigerant Leaks	Item: 3.1.3
Category: Emergency Procedures	Page: 1 of 1

Description: This describes the action to take in the event of a refrigerant leak.

Notes:

1. The refrigerant that leaks out will most probably be in the form of vapour, therefore the source of the leak will be hard to find out without proper leak detection equipment.

Step	Action	Notes
1	Shut down and electrically isolate the affected items of plant. Check that all doors of control panels are properly closed.	
2	Evacuate all personnel from the vicinity and call in Refrigeration Engineers.	There must not be any smoking or flames present as they will cause the refrigerant to break down into highly toxic gases.
3	Ensure that all doors to the room are securely shut.	
4	It should be remembered that refrigerants are harmful to the Ozone Layer and therefore must be disposed of in accordance with the requirements of the local Environmental Health Officer.	

SECTION THREE

Title: Fire	Item: 3.1.4
Category: Emergency Procedures	Page: 1 of 1

Description: This describes the action to take in the event of fire breaking out.

Notes:

1. Always inform the Fire Brigade should fire break out, and never attempt to fight it alone.
2. Never use a liquid type extinguisher to fight electrical fires.

Step	Action	Notes
1	Under most conditions occurrence of fire will be picked up by the automatic fire detection equipment. However, for certain locations the manual call points should be activated.	Check the fire alarm is genuine and not a fault.
2	Having ascertained the location of the fire, check out its severity and if local and contained, use fire fighting equipment provided - hand extinguishers, hose reels, etc.	
3	If the fire cannot be extinguished within a few minutes, the general alarm should be initiated and the FIRE BRIGADE contacted. All plant should be shut down.	The standard fire drill should be put into operation. Check that all personnel are evacuated from the immediate and surrounding areas first, then from the building completely.
4	The fire should be contained as far as possible, without personal risks taken, until the fire brigade has arrived.	Check that all windows in the immediate and surrounding areas are closed, together with all fire doors. Beware of toxic fumes when in the vicinity of fire. These are not necessarily visible. If in any doubt, abandon the attempt to fight the fire. Remember, the gases are buoyant due to heat, therefore, the air is more breathable nearer floor level.

Title: Principles and Practice	Item: 3.2.1
Category: First Aid Procedures	Page: 1 of 2

a) - **Definition**

First Aid is based on the principles of practical medicine and surgery; a knowledge of the subject in case of accident or sudden illness, enables trained persons to give such skilled assistance as will preserve life, promote recovery and prevent the injury or illness becoming worse until medical aid has been obtained.

NOTE: Medical Aid indicates treatment by a Doctor.

First Aid consists of simple measures that anyone can learn but if carried out correctly, quickly, gently and as early as possible, they can be life-saving.

b) - **First Aider's Responsibility**

This ends when handed over to the care of a Doctor, a Nurse or other responsible person, but not until they have taken over the whole responsibility for the case. The First Aider should not leave until he has made his report to the Doctor or other responsible persons and has ascertained whether he could be of any further help.

First Aid, in general, is limited to the assistance rendered at the time of the emergency, with such material as is available and often extensive improvisation will be necessary.

Remember at all times the importance of COMMON SENSE in First Aid, as an addition to the actual knowledge of the subject. This section (and other First Aid Manuals) usually considers for treatment only one condition at a time. However, in real life it will be found that serious accidents rarely produce a single injury. Frequently, two injuries or more occur close together, so that the correct treatment of one may interfere with the correct treatment of the other. One injury may require the casualty to be put on his back, another that he should be in a coma or recovery position. In such a circumstance, the First Aider must decide which injury is most serious, or needs the most urgent treatment and treat that one in the correct way and then deal with the second injury as correctly as possible under the conflicting circumstances. Often a casualty suffers from more than two injuries at the same time, which makes matters even more difficult.

c) - **The Scope of First Aid**

This consists of three parts;

- i) Dealing with the situation, apart from the casualty.
- ii) Diagnosing what is the matter with the casualty and then giving the correct First Aid treatment.
- iii) Transferring of the casualty to a doctor, hospital or home and notifying those concerned about the accident.

Title: Principles and Practice	Item: 3.2.1
Category: First Aid Procedures	Page: 2 of 2

In arriving at the diagnosis, the First Aider is guided by:

- i) The report furnished by persons present (which includes the conscious casualty), as to the cause of the injury or illness. - HISTORY
 - ii) The account given by the casualty of their own sensations and feelings. - SYMPTOMS
 - iii) The complete examination of the patient. - SIGNS
- d) - **Priority in First Aid**

Do first things first, quickly and methodically.

Check for danger.

Reassure the casualty and those around to lessen anxiety, whilst taking in the situation.

Ascertain from the casualty whether they are on any medication or have any medical conditions.

If breathing has stopped, start resuscitation.

Control visible bleeding.

Give priority to the individual with the most important injuries.

Give priority where several are injured to those who will benefit most by prompt treatment.

Guard against shock - and look for concealed bleeding.

Immobilise fractures.

Do not attempt too much - attend to the essential and prevent the condition from becoming worse.

Do not allow people to crowd around - they get in the way, and fresh air is essential.

Arrange early for careful conveyance of the casualty to Hospital or to a Doctor.

Title: Life Saving Measures	Item: 3.2.2
Category: First Aid Procedures	Page: 1 of 1

a) - **Electrical Injuries**

Even with domestic voltages, if an electric current passes through a person it may, in some cases, produce stoppage of breathing, burns and cardiac arrest.

b) - **Breathing Stopped**

If the victim stops breathing he will die unless breathing is restored at once. First tilt his head back to open the air passage from mouth to lungs, squeeze the nostrils together then blow your own breath through his mouth into his lungs. If there is no improvement, it is likely that the heart has stopped beating and must be restarted by compressing it by manual pressure in the chest wall.

c) - **Bleeding**

Bleeding from injuries must be controlled, as severe loss of blood may lead to death. The best way to stop bleeding is to squeeze the injured part together BY DIRECT PRESSURE of the fingers on the wound, or squeeze the edges of the wound together.

d) - **Unconsciousness**

The willing but untrained bystander is most helpless when confronted with the UNCONSCIOUS victim. The simplest act of turning such a victim on his side, in the RECOVERY position, so that he cannot drown in his own vomit, may save as many as 20% of such victims who would otherwise die.

e) - **Shock**

Shock is likely to be present in all cases of injury and many cases of sudden illness. Its effects, which may be extremely serious, may be mitigated by the comfort, confidence and reassurance supplied by the rescuer.

f) - **Broken Bones**

These are serious injuries - STOP ANY MOVEMENT OF BROKEN BONES which may make the injury more severe. Injured limbs may be secured to the body or the other uninjured limb.

g) - **Burns and Scalds**

These are common injuries and if a large part of the body is involved, death may result. Cool the affected area with cold water then cover with clean cloth or large dressing until seen by a Doctor.

Title: Emergency Situations	Item: 3.2.3
Category: First Aid Procedures	Page: 1 of 2

a) - **The Situation**

Be calm and take charge.

Ensure safety, i.e. from the possibility of falling masonry, etc.

Ask those present to remain, if considered responsible, as they may be able to help; otherwise they should be requested to stand clear.

Give each one a specific job,

e.g. Ring up and notify the Police.

Ask for an Ambulance or send for a Doctor.

(In each case, state the place of the accident and inform what has happened).

Ask if anyone has any First Aid knowledge.

Ask for help in turning the casualty or in steadying a limb.

(In each case, give exact instructions and if necessary show the bystander how your request should be carried out).

b) - **The Casualty**

Depending on what has happened and the degree of severity of the injuries, decide whether to treat the casualty where they are or whether to move them to a more suitable place.

If you decide to move him, carry out a quick preliminary examination of the head and neck, spine and four limbs. Then decide on the most suitable method of removal in view of the injuries and the amount of skilled and unskilled help available. Then complete the examination of the casualty for injuries so that you can make a complete diagnosis and carry out the necessary treatment.

c) - **The Transferral**

Stay with the casualty and reassure them until an Ambulance or Doctor arrives. Give your report to the Doctor and if necessary accompany the Casualty to Hospital and report there. Notify the nearest relative and any other person or organisation that should be told. In serious outdoor accidents the Police should be sent for or notified.

Title: Emergency Situations	Item: 3.2.3
Category: First Aid Procedures	Page: 2 of 2

d) - **Summary**

THE MOST URGENT MATTERS ARE:-

1. Check for danger.
2. Get or send for help in accordance with current procedures.
3. To restart the heart if it has stopped beating.
4. To apply resuscitation if the casualty is not breathing. If in doubt as to whether the casualty is alive or not, continue treatment until medical aid is available.
5. To control bleeding.
6. To maintain a clear airway by correctly positioning the casualty.

THE MOST IMPORTANT PROCEDURES TO PREVENT THE CONDITIONS BECOMING WORSE ARE:-

1. To dress wounds.
2. To immobilise fractures and to treat large wounds.
3. To place the casualty in the most comfortable position, consistent with the requirements of treatment.

THE MOST HELPFUL MEASURES IN PROMOTING RECOVERY ARE:-

1. To relieve anxiety and promote confidence.
2. To relieve pain and discomfort.
3. To protect from cold.
4. To handle gently so as to do no harm.

Title: Dressings	Item: 3.2.4
Category: First Aid Procedures	Page: 1 of 1

a) - **Purpose**

Dressings are used to:-

- i) Assist in controlling bleeding.
- ii) To protect a wound from further injury.
- iii) To prevent or lessen infection.

An efficient dressing should be non-adherent and have a high degree of porosity.

b) - **Types of Dressings**

i) **Adhesive Dressings**

The wound covering is of a non-adherent film which is supported by several layers of absorbent gauze or cellulose held in place by a covering layer of adhesive material. Sterilized dressings are sealed in a paper or plastic envelope and should not be opened before use.

ii) **Bandages**

Bandages are made from flannel domett (cotton and wool), calico, elastic net or paper. They can be improvised from any of the above materials.

There are three different types;

- a) Triangular bandage
- b) Roller bandage
- c) Elastic net

c) - **Purposes of Bandages**

- i) To retain dressings and splints in position.
- ii) To maintain direct pressure over a dressing to control bleeding.
- iii) To reduce or prevent swelling.
- iv) To afford support in the form of a sling.
- v) To act as padding when nothing suitable is available.

Title: Treatment for Electric Shock	Item: 3.2.5
Category: First Aid Procedures	Page: 1 of 1

The following are the basic steps associated with treating a person who has suffered an electric shock:-

- 1) Isolate the electrical supply.
- 2) Use a non-conducting material, such as wood, to pull the victim away from contact with live parts or wires.
- 3) If the victim is not breathing, administer cardiopulmonary resuscitation (CPR) after breaking contact with the electrical source.
- 4) Call a physician and continue CPR until breathing starts or until a physician arrives.
- 5) Treat an electrical burn as a thermal burn by applying clean, cold (iced) compresses. Prevent contamination, and cover with a clean, dry dressing.

SECTION THREE

Title: Further Reading	Item: 3.2.6
Category: First Aid Procedures	Page: 1 of 1

The following books, published by St. John Ambulance Association, to which this section is indebted, are strongly recommended for further information.

- 1) Digest of First Aid.
- 2) Authorised First Aid Manual.

Both of the above publications can be obtained from:-

St. John Ambulance Association and Brigade
1, Grosvenor Crescent,
London, SW1

SECTION FOUR

HEALTH AND SAFETY INFORMATION

INDEX

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Title: Introduction	Item: 4.1.1
Category: General	Page: 1 of 1

This manual has been compiled within the framework of the Construction (Design and Management) Regulations 1994, which place specific duties on Clients, Designers and Contractors to co-ordinate and manage effectively the health and safety of workers and others throughout all stages of a construction project from conception, design and planning through to the execution of works on site and subsequent maintenance and repair.

It is intended that this manual forms part of the general Health and Safety file, the purpose of which is to alert those responsible for the structure, after handover, to the health and safety risks that must be managed when the structure and associated plant is maintained, repaired, renovated or demolished.

Health and safety issues are important in maintenance because statistics show the maintenance worker to be at greater risk to accident and injury. This is partly because these workers are exposed to more hazards than others, and partly because there are pressures of time and money upon the maintenance task to be completed as quickly as possible. Less thought is given to the special needs for health and safety of the maintenance task than is given to routine tasks which are easier to identify, plan and control.

Title: Maintenance Policy	Item: 4.1.2
Category: General	Page: 1 of 1

From the health and safety aspect, defects requiring maintenance attention, which have led, or could lead to, increased risk to the workforce, should receive a high priority. Linking inspections with maintenance can be useful so that work areas and equipment are checked regularly for present and possible future defects. Some plant items are subject to statutory maintenance requirements as well as the manufacturers requirements. These should all be strictly adhered to.

The need for maintenance of each piece of equipment has been anticipated in its particular design. The overall maintenance of the complete system has been made safer by its consideration at the design and later installation stage.

This manual contains all the information required to enable preventative and, when necessary, breakdown maintenance being carried out, i.e. what action needs to be taken and how often. This particular section contains details on good practice and special checks and procedures related to health and safety.

Title: Causes and Avoidances of Accidents	Item: 4.1.3
Category: General	Page: 1 of 1

Maintenance accidents are generally caused by one, or more, of the following factors:-

- i) Lack of perception of risk by managers/supervisors, often because of lack of necessary training.
- ii) Unsafe, or no systems of work devised, e.g no permit to work systems in operation, no facility to lock off machinery and electrics before work starts and until work has finished.
- iii) No co-ordination between workers or communication with other supervisors or managers.
- iv) Lack of perception of risk by workers, including the failure to wear protective clothing or equipment, no co-ordination,
- v) Use of maintenance contractors who are inadequately briefed on the health and safety aspects and not selected on health and safety as well as cost grounds.

The safe operation and maintenance of systems requires steps to be taken to control the causes of accidents. These steps can be divided into the following phases:

- i) Planning - Identification of the need for planned maintenance and arranging a programme for this.
- ii) Evaluation - The hazards associated with each maintenance task should be listed and the risks of each considered (frequency of the task and possible consequences of failure to carry it out correctly). The tasks can then be graded and the appropriate degree of management control applied to each.
- iii) Control - The control for each task will take the above factors into account, and will include any necessary review of design and installation, training, introduction of written safe working procedures to minimize risk, i.e. a safe system of work, allocation of supervisory responsibilities and necessary allocation of finances. Review of the activities of contractors engaged in maintenance work will be required in addition.
- iv) Monitoring - Random checks, safety audits and inspections and the analysis of any reported accidents for cause, which might trigger a review of procedures, constitute necessary monitoring to ensure the control system is fully up to date. The introduction of any change in the workplace may have maintenance implications and should therefore be included in the monitoring process.

Title: Introduction	Item: 4.2.1
Category: Good Practice	Page: 1 of 1

It is the prudent duty of the employer to ensure, as far as is reasonably practicable, the health and safety of the maintenance staff and other people using the premises. This section of the manual is a guide to some of the precautions to be taken by both management and staff to achieve this.

The equipment and materials referred to in this section should be used, operated and maintained strictly in accordance with the operating instructions or manufacturers' literature as appropriate.

It is the duty of every member of the maintenance team to take reasonable care for their own health and safety and that of other people, who may be affected by their actions or omissions whilst at work. It is strongly recommended that at least one copy of the Health and Safety at Work Act (1974) is available on the premises and has been read and understood by all maintenance personnel.

Maintenance staff should ensure that they are aware of and understand the "IN HOUSE" rules and regulations which have been formulated and will incorporate safety precautions found to be necessary for local conditions.

Title: Safe Working Conditions	Item: 4.2.2
Category: Good Practice	Page: 1 of 1

Work should not be carried out under conditions where there is an element of danger, if it is reasonably practical to eliminate the risk. Safe working conditions should be provided to give protection and safe ingress/egress where necessary. Work on live electrical circuits is not allowed.

Access doors and panels to all areas containing plant e.g. plant rooms, electrical panels, fan chambers etc., should be suitably identified and where appropriate carry warning notices.

Adequate ventilation should be provided. Special precautions should be taken where steam, smoke or any other unhealthy or dangerous contaminants are present, particularly in confined spaces.

Lighting must be maintained at a high level in all plant rooms and service areas to enable inspections to be carried out at any time, day or night and to avoid the risk of accident.

A high degree of cleanliness, both of rotating machinery and static equipment, is of the utmost importance. Floors and machinery must be protected from spilt lubricants and all loose materials, containers and papers must not be allowed to accumulate.

Title: Safe Way to Work	Item: 4.2.3
Category: Good Practice	Page: 1 of 1

This can be summarised by a procedure which can be adopted for each maintenance risk.

Think about the dangers:

- Check the task and site for hazards.
- Decide on the precautions needed.

Prepare:

- Put on safety clothing.
- Assemble the tools and equipment.
- Position **DAI GER** and **WARI II G** notices.
- Erect safety barriers.
- Position equipment in a safe place.

Carry out task:

- Work in a logical sequence.
- Use the correct tool for the job.
- Never improvise on tools.
- Discard defective tools.

Clear up after completion of task:

- Collect all tools and equipment and return to store.
- Remove notices or safety barriers.
- Tidy up work area.

Title: Personal Safety	Item: 4.2.4
Category: Good Practice	Page: 1 of 1

The maintenance personnel must always use the protective clothing available. These items are the normal requirement for the maintenance task.

<i>Overalls or Dust Coats</i>	These must be worn at all times to protect the maintenance personnel from dirt and spillage.
<i>Gloves</i>	Reinforced non-slip, heavy duty industrial gloves must be worn when handling heavy equipment, dirty or greasy items, or sharp edged metal parts.
<i>Rubber Boots</i>	Must be worn when harmful chemicals are being handled, for example stand-by battery supplies.
<i>Face masks</i>	These must be worn when handling harmful chemicals.
<i>Safety Shoes or Boots</i>	Must be worn at all times. The reinforced sole and toe caps protect the feet when walking on rough ground or moving heavy equipment.
<i>Safety Glasses</i>	These must be worn to prevent eye injuries, for example cutting wire or cleaning sand and dust from equipment.
<i>Ear Protection</i>	Must be worn in areas of high noise intensity, for example near diesel generators or air compressors. The two main types are ear muffs and earplugs.
<i>Safety Helmet</i>	Must be worn at all times when working in the site areas, underneath any equipment, scaffolding or pipe runs or any other overhead hazard. It should be noted that all safety helmets carry an expiry date and should not be used after this time.

Title: Permit to Work	Item: 4.2.5
Category: Good Practice	Page: 1 of 3

A "Permit to Work" system must be adopted to meet and satisfy the legal requirements of the present legislation, i.e. "Health and Safety at Work Act (1974)".

The "Permit to Work" system should incorporate the provisions listed below:

- i) Before working on any items of equipment a "Permit to Work" docket, signed by an appropriate authority, must be obtained. On completion of the work item, it should be returned for "logging" in the master maintenance systems records.
- ii) To give authority to maintenance staff to commence inspection of plant, or to work on plant.
- iii) To explain in outline the approach required to carry out the work in such a way that there is no potential hazard to personnel or plant.
- iv) The system should be devised so that maintenance may be carried out safely and that no starting up, or running, of plant presents an environmental hazard.
- v) Co-ordination of staff requirements in relation to maintenance procedures and operations to eliminate exposure of the work force to any hazard.
- vi) Correct shutting down of systems must be implemented and no electrical work on equipment where removal of guards or housings has taken place, must be undertaken without satisfactory clearance that the plant is now safe for work to be proceeded with.
- vii) Members of the maintenance staff must only carry out work within their discipline and job description.
- viii) Where it is necessary to provide staging, scaffolding, towers and ladders to examine or carry out replacement of components, then these must comply in all respects with the latest legislation.
- ix) The authorized person will:-
 - a) Make sure all safety precautions are taken and recorded.
 - b) Enter on the work permit:-
 - What work has to be done.
 - What piece of equipment has to be made safe.
 - What safety precautions have been taken.

Title: Permit to Work	Item: 4.2.5
Category: Good Practice	Page: 2 of 3

- x) Maintenance tasks require strict adherence to safe working practices for the following reasons:-
 - a) There will be a number of safety precautions.
 - b) Equipment close by may still be operating.

Notes on Permit to Work

Part 1 should be completed by the person authorising the work, before issuing to the person for the switching, isolation, testing and earthing operations. Where the person authorising the work is responsible for the switching etc., they should do this for themselves as a check on procedure.

Part 2 should be completed by the person responsible for the switching, isolating, testing and earthing operations after their completion.

Part 3 both the original and duplicate should be signed by the person actually in charge of the work on the apparatus. The original should be retained by the person actually in charge of the work until the work is stopped or completed. The duplicate should be retained by the person authorising the work.

Part 4 should be signed by the person actually in charge of the work when the work has been stopped, or completed, and then handed back to the person who authorised the work.

Part 5 should be signed by the person responsible for signing Part 2, or in their unavoidable absence, by a person of equivalent status and understanding.

SECTION FOUR

Health and Safety Information

Title: Permit to Work	Item: 4.2.5
Category: Good Practice	Page: 3 of 3

Example of Permit to Work

Permit to Work on Electrical Apparatus	I O.....
Part 1 Authorization for Work.	
A. This permit is issued for the following work	
By To	
being an authorised person in charge of switching and earthing operations.	
B. Switching and Isolating. The apparatus to be disconnected from all live conductors by the following operation	
C. Potential Tests and Earthing. Potential indicator test to be applied before fixing earths at the following points .	
D. Danger Notices are to be posted at:-	
E. Caution Notices are to be posted at:-	
Time	Date Signed Rank

Part 2 Declaration
I hereby declare that the operations in Part 1, B, C, D & E have been completed and apparatus detailed in Part 1, A, is safe to work on, and any adjacent live equipment is securely padlocked. It will not again be made "live" until the Clearance Certificate in Part 4 is completed.
ALL OTHER APPARATUS IS DANGEROUS
Time Date Signed Rank

Part 3 Receipt
I hereby declare that I accept responsibility for doing the work on the apparatus detailed in Part 1, A, of this Permit and that no attempt will be made by me or by men under my control to work on any other apparatus.
Time Date Signed Rank

Part 4 Clearance of Work
This is to certify that the work detailed in Part 1, A above has been (stopped/completed)* men, gear and tools withdrawn, local earths removed.
Time Date Signed Rank
(Delete the word that does not apply.)

Part 5 Cancellation
I hereby declare this Permit to Work cancelled, and the earth connections applied at points indicated in Part 1, C, have been removed. The apparatus is now ready for switching into service.
Time Date Signed Rank

Title: Warning and Safety Notices	Item: 4.2.6
Category: Good Practice	Page: 1 of 1

The "The Health and Safety (Safety Signs and Signals) Legislation 1996" make it a legal requirement that all safety signs comply with BS.5378: Part 1, 1980. BS.5378: Part 1, 1980 divides safety signs and notices into four categories, according to the type of "Message". Each category has a distinctive colour and shape as summarised below.

Sign Type	Colour	Shape	Meaning
Prohibition	Red	Outline circle with cross bar	a) You must not b) Do not do c) Stop
Mandatory	Blue	Solid circle	a) You must do b) Carry out the action given by the sign
Warning	Yellow	Outline triangle	a) Caution b) Risk of danger c) Hazard ahead
Safe Condition	Green	Rectangle	a) The safe way b) Where to go in an emergency

The following recommendations should be observed:-

- i) At all times warning and safety notices must be maintained and properly displayed and worded to give clear instructions.
- ii) Notices covering First Aid and Resuscitation from Electric Shock must be displayed in all plant rooms.
- iii) Proper fire precautions must be observed within the plant rooms and building.
- iv) Notices must be displayed stating:-

! O SMOKI! G or SMOKI! G PROHIBITED !! THIS AREA

- v) All fire fighting equipment must be clearly marked and available for use.

Title: General	Item: 4.2.7
Category: Good Practice	Page: 1 of 1

a) **Escape Routes**

- i) Periodic inspection of all escape routes, fire doors, fire exits., should be made to confirm that doors open freely and routes are always kept clear of obstructions.
- ii) Should oil be spilt on bases or floors, the area must be thoroughly cleaned and sanded.

b) **Keys**

Access to all plant rooms, electrical cupboards and equipment therein, which require a key to open for access, should be controlled and restricted to authorised personnel only.

c) **First Aid**

First Aid boxes should be provided in accessible positions. These should be kept clean and in good repair. Contents of the boxes should be regularly checked and missing items replaced.

d) **Protective Clothing**

Where appropriate, wear protective clothing and equipment, i.e. goggles, respirators, safety harnesses. Protective screens should be provided when arc welding is in progress.

e) **Drive Guards and Housings**

Never work on moving parts of machinery. Exercise caution on the stationary parts. Ensure that moving parts are protected by suitable guards. Be aware of warning notices such as:-

"Remotely Controlled Plant"

Do not put hands or objects on any item of stationary rotating machinery as this may start automatically at any time.

WARNING:- Before commencing maintenance work on any item of plant, ensure the plant is mechanically isolated from the system and that the electrical supply has also been isolated and locked off. Ensure that all control and ancillary electrical circuits have been made safe.
--

f) **Replacement Parts**

All safety procedures are to be observed when using replacement parts, all electrical circuit diagrams, manufacturers' spares lists must be checked against the existing equipment to determine it is the correct arrangement or equipment for the specific application. Any item used must comply with the correct British Standard Specification and where tests are carried out, they must be in compliance with the relevant British Standard Specification or Code of Practice.

Title: Introduction	Item: 4.3.1
Category: Safety Procedures and Checks	Page: 1 of 1

The purpose of this list of safety procedures, is to provide general safety precautions to be used by persons who own, install, operate, or maintain the systems described within this manual.

This list should not be used to replace instructions that are provided by the equipment manufacturer. Therefore, anyone attempting to work on any equipment, should be thoroughly familiar with the specific instructions for that particular unit.

The following is a list of the criteria that have been used here to indicate the intensity of the hazard:

- | | |
|-------------------------------|---|
| DAJ GER | - This means that there is an immediate hazard that will result in severe personal injury or death. |
| WARJ IJ G | - This means that hazards or unsafe practices could result in severe personal injury or death. |
| CAUTION | - This means that potential hazards or unsafe practices could result in minor personal injury. |
| SAFETY IJ INSTRUCTIONS | - These are general instructions that are necessary for safe working practices. |

Title: Personal Protection	Item: 4.3.2
Category: Safety Procedures and Checks	Page: 1 of 1

Warning

1. Do NOT touch electrical wiring connections without first isolating the source of supply.
2. Do NOT touch electrical wiring connections with wet hands.
3. Do NOT touch electrical equipment while standing on a wet surface or wearing wet shoes.
4. Do wear a hard hat or other protection, when there is a possibility of falling objects.

Caution

1. Do wear safety glasses equipped with side shields when working in workshops or at construction sites.
2. Do wear gloves when handling system components after a compressor motor burnout. The refrigerant and oil contain acid that can result in acid burns to the skin.
3. Do wear goggles and gloves when handling chemicals; when welding, cutting, grinding or brazing; or when in an area where these operations are performed.
4. Do wear gloves and other protective clothing when working with sheet metal.
5. Do wear safety shoes when working around or lifting heavy objects.
6. Do wear protective clothing when arc welding to protect from serious burns.
7. Do wear ear protection when working in areas where sound levels are greater than 90db.
8. Do NOT wear jewellery, loose clothing, long ties, or gloves whilst working around moving belts and machinery.
9. Do NOT wear rings or watches whilst working on electrical equipment.

Safety Instructions

1. Keep your work area clean of debris and free of liquid spills.
2. Do NOT continue working if you become ill. An unwell person is less observant and is therefore more prone to accidents.

Title: Working with Electricity	Item: 4.3.3
Category: Safety Procedures and Checks	Page: 1 of 1

Danger

1. Do NOT attempt to check voltage without proper instructions.

Warning

1. Do NOT work on electrical circuits until they are opened, tagged, and locked.
2. Do NOT attempt to stop a unit by opening an isolating or fuse switch.
3. Do NOT attempt to check the resistance on an energised circuit.
4. Do NOT remove terminal box covers while a motor is running.
5. Do NOT handle capacitors until they have been properly discharged.
6. Do earth all electrical equipment.
7. Do use an earth fault current interrupter when using hand tools. (ie. RCD or RCBO).

Caution

1. Do NOT bypass or block electrical circuit interlocks.
2. Do NOT check electrical equipment until you are certain that the electrical power is off in all circuits.
3. Do use insulated fuse pullers on cartridge fuses.

Safety Instructions

1. Do NOT store tools or parts in control or starter boxes.
2. Do NOT use meters for purposes other than those for which they have been designed.
3. Do periodically inspect electrical flex and replace, or repair, any damaged flex, plugs or connections.

Title: Machinery	Item: 4.3.4
Category: Safety Procedures and Checks	Page: 1 of 2

Before any machinery is worked on, it is essential that it is isolated and cannot be accidentally set in motion. Procedures to ensure this are:-

- a) All isolation switches or circuit breakers are locked in the OFF position.
- b) Warning notices stating:

MACHINERY UNDER REPAIR DO NOT SWITCH ON

displayed at the machine.

- c) For major test and inspection the power supply must be disconnected.

Where ever possible the maintenance tasks must be performed on stationary machinery. In the exceptional circumstances that this is not possible, **GREAT CARE** must be taken to avoid injuries and work must not be carried out alone.

All live electrical terminals must be identified with warning notices stating:-

DAANGER - LIVE ELECTRICAL PARTS

All guards, screens, shields, covers and lids must be secured with all screws or nuts and bolts, and warning notices removed before the machinery is operated.

Guards - Types

<i>Fixed Guards</i>	Permanently in position to prevent access to the dangerous parts from all directions.
<i>Interlocked Guards</i>	Designed to remain in the closed position whilst machine is in motion.
<i>Automatic Guards</i>	Designed to remove the operator from danger.
<i>Trip Guards</i>	Designed to stop the machine in case of danger.

SECTION FOUR

Title: Machinery	Item: 4.3.4
Category: Safety Procedures and Checks	Page: 2 of 2

Checks

Are all dangerous parts guarded?

Are all guards in good condition and properly secured?

Does the guard prevent the operator, or anyone else, from reaching over or under into the dangerous part?

Are all the guards working properly?

Are the work feed openings of fixed guards small enough to prevent the operator from reaching through to a dangerous part?

Are the start and stop controls easily accessible, in working order, and clearly identifiable?

Can the machine be positively isolated electrically?

How frequently are the guards or safety devices inspected or tested?

Title: Tools and Clothing	Item: 4.3.5
Category: Safety Procedures and Checks	Page: 1 of 1

The maintenance personnel must be prepared to go to any part of the site to carry out a task. It is not always possible to take a complete tool kit, so careful forward planning is required. An incorrect tool must never be used. The maintenance personnel must keep their tools in good working order. Any tool showing signs of wear, damage or defect must be replaced.

Checks

Are the correct tools being used?

Are the handles secure?

Are the chisels and punches mushroom headed? (They should be ground smooth).

Are spanners being abused by the use of extension tubes?

Worn or defective tools should be disposed of and replaced.

Does all protective clothing fit correctly?

Do new employees have to wait for an issue of protective clothing?

Are visitors issued with protective equipment?

Title: Lifting and Access Methods	Item: 4.3.6
Category: Safety Procedures and Checks	Page: 1 of 1

To gain access to equipment it is often necessary for maintenance personnel to remove guards, screens, shields, covers and lids.

All screws or nuts and bolts must be kept safe.

To gain access to equipment installed in difficult locations, maintenance personnel often need to erect access equipment.

The access equipment must be safe and secure, footholds and handholds must be provided if the maintenance person is liable to fall more than two metres. This includes chains, ropes, slings, hooks, eye bolts and shackles.

Checks

Is the lifting tackle being used correctly? Is the safe working load clearly marked?

Is the equipment being examined every six months?

Is there a certificate of test showing the safe working load?

Is the ladder secured at the top? If not, is it firmly fixed at the bottom or footed by a second person?

Is the ladder erected with the appropriate height/distance from wall ratio?

Is the ladder in good condition, no missing or improvised rungs?

How frequently are the ladders inspected?

Title: Good Housekeeping	Item: 4.3.7
Category: Safety Procedures and Checks	Page: 1 of 1

Maintenance work is not always possible under ideal conditions, for example, working above ground or in confined spaces. It is therefore essential that a tidy area, clear of obstructions and loose tools, is kept during the maintenance task.

Checks

Is there a place for everything and is everything in its place?

Are gangways, aisles, passageways and stairs kept clear and free from substances likely to cause a person to slip?

Is the lighting satisfactory?

Are handrails provided on stairs?

What is the system for cleaning up spillage of materials or substances?

Are all floors and stairs in good repair?

Are potentially harmful substances used in your workplace, if so:-

- i) Are they clearly marked with instructions as to how they should be used, stored and disposed of?
- ii) Are the employees trained in their use?
- iii) Is the workplace adequately ventilated?
- iv) What is the system for dealing with an emergency, in relation to those substances?

Hot, Scalding or Corrosive liquids may be used in the workplace:-

- i) Are the tanks containing these liquids, fenced to a height of 4 metres or securely covered?
- ii) Are any gangways above the tanks provided with adequate guard rails?

Have all Control of Substances Hazardous to Health (COSHH) requirements been met?

Title: Fire Precautions	Item: 4.3.8
Category: Safety Procedures and Checks	Page: 1 of 1

Are the quantities of highly flammable liquids stored in the workroom, kept to a minimum?

Are the empty containers and contaminated rags etc., removed immediately from the work place?

Are there any naked flames or any other possible sources of ignition present when using flammable liquids?

Is smoking prohibited, and if so, is the rule being observed?

Are all the extinguishers in their correct position, full and free from obstruction?

Are all the gangways and staircases which may be used as a means of escape, free from obstruction?

Can every door along the escape route be easily and immediately opened from inside?

Have fire resisting or smoke stop doors been wedged open?

Can the Fire Alarm be heard in all parts of the work place?

Has the fire alarm been tested during the last three months?

When was the last fire evacuation drill?

What potentially hazardous dusts are used in your work place?

What safeguards are used to minimise risk?

SECTION FIVE

OPERATING PROCEDURES

INDEX

<u>ITEM</u>	<u>DESCRIPTION</u>
5.1	- <u>GENERAL</u>
5.1.1	- Introduction
5.2	- <u>FILLING PROCEDURES</u>
5.2.1	- Direct Expansion Air Conditioning Systems
5.3	- <u>DRAINING PROCEDURES</u>
5.3.1	- Direct Expansion Air Conditioning Systems

Title: Introduction	Item: 5.1.1
Category: General	Page: 1 of 1

The procedures in this section are laid out on a step-by-step basis. No step within the procedure should be taken unless the preceding step has been satisfactorily completed, unless otherwise stated. Before carrying out any procedure it should be read through to ensure that it is completely understood and that any tools, equipment, spares etc., required, are to hand. Additionally, the relevant descriptions and plant data should be read for familiarisation of the particular system to which the procedure refers.

The building services have been generally designed for automatic operation. The systems are inter-dependant on each other and, therefore, should not be treated in isolation in the normal operation of the building.

Title: Direct Expansion Air Conditioning Systems	Item: 5.2.1
Category: Filling Procedure	Page: 1 of 1

Description: This describes the filling procedure for a direct expansion air conditioning system.

Notes:

1. The following works should only be carried out by a qualified refrigeration engineer using the correct items of equipment.

Step	Action	Notes
1	Carry out a pressure test of the complete refrigeration pipework system.	Refer to the manufacturers' literature for pressure limits.
2	Upon completion of a pressure test, release the pressure very carefully and remove test equipment.	Care should be taken when releasing the high pressures involved.
3	Fit evacuation equipment to the pipework and evacuate the lines to remove all air, condensation etc.	
4	During the evacuation period check over the equipment to ensure all components are ready to operate.	
5	Remove evacuation equipment and fit refrigerant gas canister leads to pipework. Switch on main electrical supply and begin the filling of the system.	Refer to manufacturers' literature for the quantity of gas to be used.
6	Upon completion of the filling operation remove the gas canister leads and check the operation of all equipment and controls.	
7	Ensure that crankcase heater is functional and leave system for a minimum of eight hours before putting into operation.	This period allows the heater to reach its operating temperature and the gas to settle.
8	Finally check the function of all controls.	The system is now ready to operate under the dictates of the controls system.

Title: Direct Expansion Air Conditioning System	Item: 5.3.1
Category: Draining Procedure	Page: 1 of 1

Description: This describes the procedure for draining a direct expansion air conditioning system.

Notes:

1. The following works should only be carried out by a qualified refrigeration engineer using the correct items of equipment.

Step	Action	Notes
1	Switch off controls operation.	
2	Fit refrigerant gas collection canister leads to the pipework and draw off the refrigerant.	All refrigerant must, by law, be removed for proper recycling or disposal if contaminated.
3	Upon completion of the degassing process, disconnect leads, close valves and remove collection unit.	
4	Switch off local isolators and main electrical supply.	This ensures that the crankcase heater is taken off line.

SECTION SIX

MAI TEI AI CE SCHEDULES

INDEX

<u>ITEM</u>	<u>DESCRIPTION</u>
6.1	- <u>GENERAL</u>
6.1.1	- Introduction
6.2	- <u>COOLING SYSTEMS</u>
6.2.1	- Direct Expansion Air Conditioning Unit
6.2.2	- Compressor
6.2.3	- Filter (Panel Washable)
6.2.4	- Cooling Coil (DX)
6.2.5	- Fan
6.2.6	- Small Fan
6.2.7	- Electric Motor
6.2.8	- Insulation

Title: Introduction	Item: 6.1.1
Category: General	Page: 1 of 1

Only a regular and organised scheme of maintenance work, planned, to cover all details of the installation, within given maintenance periods, can ensure continued satisfactory operation with a minimum liability to interruptions to supply due to equipment faults. A planned maintenance scheme should include a system of logging, wherein records are kept of inspection, maintenance and repair of all items of plant and equipment.

Careful attention must be given to securing the safety of personnel and equipment while maintenance or repair work is in progress. A code of safety rules based on a system of "permits to work" is recommended. Plant rooms should be kept clean and the practice of using plant room space for storage must not be permitted. Access to plant rooms and equipment must be limited to authorised personnel who are members of the engineering staff.

Where maintenance work is in progress, a DANGER notice must always be attached to any "live" apparatus calling attention to the danger of approach. A CAUTION notice must always be attached to plant or its associated control equipment which may be occasioned by interference. Before any work is commenced on any item of equipment, the supply and ancillary circuits must be isolated and locked off if possible.

All safety precautions and general recommended procedures relating to the maintenance of switchgear are contained in the British Standard CP5405 (1976) and this should be read in conjunction with the instructions for routine periodic maintenance given in the following pages.

SECTION SIX**Maintenance Schedules**

Title: Direct Expansion Air Conditioning Unit	Item: 6.2.1
Category: Cooling Systems	Page: 1 of 1

Item	Frequency	Details	Notes
Internal Unit			
1	Monthly	Remove and clean return air filter.	Brush clean or wash in warm soapy water.
2	3 Monthly	Clean surfaces of casing.	
3		Brush or vacuum clean evaporator coil surfaces.	
4		Check for correct operation of controls.	
5		Check fan is free to rotate.	
6		Check for unusual noise or vibration.	
7		Clean out drip tray and ensure condensate drain is free from obstruction.	
8		Check unit for signs of leakage.	
9	Various	Check and maintain component items - refer to individual component's maintenance schedule.	Small Fan, Coil, Filter (Washable Panel)
External Unit			
1	3 Monthly	Inspect unit for signs of leakage or corrosion.	Clean off any deposits and treat any bare metal accordingly.
2		Clean condenser coil surfaces.	
3		Check fans are free to rotate and impellers are secure onto shaft.	
4		Check operation of all controls.	
5		Check refrigerant charge.	Top up as required.
6		Check operation of crankcase heater.	
7		Check crankcase oil level.	Top up as required.
8		Check for unusual noise or vibration.	
9	Various	Check and maintain component items - refer to individual component's maintenance schedule.	Fan, Coil, Compressor

SECTION SIX**Maintenance Schedules**

Title: Compressor	Item: 6.2.2
Category: Cooling Systems	Page: 1 of 1

Item	Frequency	Details	Notes
1	Weekly	Check compressor for signs of leakage.	This can be viewed through the sight glass. Top up as required.
2		Check oil level.	
3		Check if oil is contaminated with refrigerant.	Excessive foaming will indicate the presence of refrigerant. Switch off unit and investigate cause.
4		Check for unusual noise or vibration.	Internal components may be broken. Overhaul or replace as necessary.
5	Yearly	Inspect compressor for signs of corrosion.	Remove rust etc and treat any bare metal accordingly.
6		Check security of electrical connections.	
7	Various	Check and maintain component items - refer to individual component's maintenance schedule.	Electric Motors.

SECTION SIX**Maintenance Schedules**

Title: Filter (Panel Washable)	Item: 6.2.3
Category: Cooling Systems	Page: 1 of 1

Item	Frequency	Details	Notes
1	Monthly	Take manometer reading. If reading shows the filter cells are at their maximum pressure drop they should be washed. After washing check condition of filter medium, if damaged replace.	Refer to the Plant Data. Check cells are replaced in the correct direction for air flow.
2		Wipe any accumulated dirt from the retaining channels and frames.	On replacing access doors check that the integral seals give a good air seal. If not, replace seals.

SECTION SIX**Maintenance Schedules**

Title: Cooling Coil (DX)	Item: 6.2.4
Category: Cooling Systems	Page: 1 of 1

Item	Frequency	Details	Notes
1	Monthly	Check pipework connections for leaks.	
2	3 Monthly	Check coil matrix is clean, unobstructed and air flow is evenly distributed over coils.	
3		Check condensate drain pans and connections are clear. Check drain pipe is properly attached.	
4	6 Monthly	Clean coil matrix.	Use soft brush and mild detergent solution. If heavily contaminated with grease use steam lance. Take care not to damage or distort fins.
5	Yearly	Check air pressure drop across coil.	Refer to the Plant Data.

SECTION SIX**Maintenance Schedules**

Title: Fan	Item: 6.2.5
Category: Cooling Systems	Page: 1 of 1

Item	Frequency	Details	Notes
1	Daily	Check fan is running smoothly without excessive noise or vibration.	
2	Weekly	Check flexible connections are secure, taut and airtight.	Connections must not resist air flow.
3		Check that fan guard is clean and unobstructed.	
4	Monthly	Check anti-vibration mountings/hangers are secure, free to move and in good condition.	
5		Check anti-back draught damper operation (air tightness and freedom of movement).	
6	6 Monthly	Check impeller and shaft for freedom of movement and balance.	
7		Check impeller and blades. Check security of impeller on shaft and inspect blades for security and damage.	
8		Check performance of fan (shaft speed, air velocity and pressure).	Refer to the certificates and the Plant Data.
9	Yearly	Clean and check casing internally and externally, repairing any corrosion or damage to finish found. Check all bolted joints are secure.	
10	Various	Check and maintain component items - refer to individual component's maintenance schedule.	Electric Motor

SECTION SIX**Maintenance Schedules**

Title: Small Fan	Item: 6.2.6
Category: Cooling Systems	Page: 1 of 1

Item	Frequency	Details	Notes
1	3 Monthly	Clean intake and discharge grilles.	Brush off any debris and wipe with a damp cloth.
2	6 Monthly	Remove grille and motor/impeller assembly and clean inner surfaces of casing.	
3		Vacuum clean motor/impeller assembly. Refit motor/impeller assembly and grille.	
4	Yearly	Remove grille and motor/impeller assembly. Strip down motor/impeller assembly and clean all items.	Wash all non-electrical components in warm soapy water. Ensure all items are dried thoroughly.
5		Apply a few drops of mineral oil to motor bearings. Refit motor/impeller assembly and grille.	
6	Various	Check and maintain component items - refer to individual component's maintenance schedule.	Electric Motor

SECTION SIX

Maintenance Schedules

Title: Electric Motor	Item: 6.2.7
Category: Cooling Systems	Page: 1 of 1

Maintenance Notes

- (a) Always ensure the electrical supply to the motor is **positively** isolated before carrying out any work. **Do not** rely on a local lockstop button for isolation.
- (b) Unless specifically designed for it, electric motors are extremely susceptible to the effect of extremes of temperature, dust and moisture. Every care must be taken to ensure they operate in a clean and dry environment.

Item	Frequency	Details	Notes
1	Monthly	Generally check motor to ensure casing ventilation louvres are unobstructed, alignment is correct, holding down bolts and pulley/couplings are secure, and motor runs without excessive vibration, temperature or noise.	
2		<i>Totally enclosed motors only</i> - Check that casing drain holes are unobstructed or if plugged, remove plugs to drain off any accumulated condensation.	
3	3 Monthly	Check motor terminals are secure, cables show no signs of overheating, terminal box is internally dry and clean, and cover is refitted securely.	Positively isolate motor before carrying out this check.
4		<i>Intermittently running motors subjected to damp conditions</i> - Check that insulation resistance across winding is at an acceptable value.	
5		<i>Single phase motors only</i> - Check centrifugal switch for wear and action.	
6	6 Monthly	Check drawn full load current does not exceed value stated on name plate and as detailed in the Plant Data.	Note measured current drawn for future reference and comparison.
7	3 Yearly	Dismantle and overhaul motor in accordance with manufacturers recommendations.	This should be carried out by a specialist.

SECTION SIX

Maintenance Schedules

Title: Insulation	Item: 6.2.8
Category: Cooling Systems	Page: 1 of 1

Item	Frequency	Details	Notes
1	6 Monthly	Inspect outer surfaces for damage. Repair or replace as necessary.	
2	Yearly	Check integrity of vapour seals.	

SECTION SEVEN

MAIN THEMES AND STRUCTURES AND LOG BOOKS

INDEX

<u>ITEM</u>	<u>DESCRIPTION</u>
7.1	- <u>INTRODUCTION</u>
7.1.1	- Formulation of a Planned Maintenance Programme
7.1.2	- Energy Usage Log Books
7.1.3	- Maintenance Log Books
7.2	- <u>RISK ASSESSMENTS</u>
7.2.1	- Working Alone
7.2.2	- Working in Occupied Premises
7.2.3	- Working in Confined Spaces
7.2.4	- Working at Heights
7.2.5	- Use of Ladders
7.2.6	- Use of Trestles
7.2.7	- Use of Hand Tools
7.3	- <u>C.O.S.H.H. INSTRUCTIONS</u>
7.3.1	- Introduction
7.3.2	- Refrigerant Gas
7.3.3	- Foam Insulation
7.4	- <u>SAMPLE LOG BOOKS</u>
7.4.1	- General Maintenance Log Book

Title: Formulation of a Planned Maintenance Programme	Item: 7.1.1
Category: Introduction	Page: 1 of 1

This Operating and Maintenance Manual is designed for use by "In-House" maintenance personnel and it is recommended that a Planned Maintenance Programme is formulated to ensure the safe and efficient operation of the plant and systems installed.

It is also recommended that a log book of any preventative maintenance tasks, faults and subsequent corrective maintenance tasks carried out, is kept. This will provide any new maintenance personnel with a history of the systems installed and highlight key areas that may require attention.

The following schedules incorporate two types of maintenance, firstly the high frequency items are mainly checks to ensure that the plant and systems are operating correctly. Secondly, the lower frequency items mainly consist of preventative maintenance tasks to ensure that the plant and system continue to operate in a safe and efficient manner.

In the formulation of a Planned Maintenance Programme, it is important to identify those items which are simply checks and those which are preventative maintenance tasks. Although each individual maintenance schedule needs to be read and understood before this can be determined, a general rule (which will not apply in all circumstances) can be applied. All items with a frequency greater than one month are checks and any items with a frequency less than one month are maintenance tasks. Items with a frequency of one month will vary from one item of equipment to the next and the general rule cannot be applied here.

Although the frequency of the preventative maintenance task is specified, the careful scheduling of the maintenance needs to be made to minimise any inconvenience caused to the building occupier.

Title: Energy Usage Log Books	Item: 7.1.2
Category: Introduction	Page: 1 of 1

The effective implementation of a planned preventative maintenance regime, as outlined within previous sections of this document, would be incomplete without the monitoring of energy usage as outlined within the Building Regulations, Part L2, Conservation of Fuel and Power (2006 Edition).

As detailed within the above regulation, suitable metering for this building will have been included within the systems at the design conception stage, together with estimated normal energy usage calculations. This information, updated to incorporate changes following system alteration, should be used as a baseline comparison to the continual log of energy usage. Any discrepancies found, either higher or lower than the estimated energy usage, should be investigated and if necessary remedial action taken. For example, higher than normal usage could indicate, leaks, insulation damage or faulty equipment.

NOTE: It must be remembered that such comparisons of baseline figures to actual readings have little meaning with regard to day-to-day usage. Only average figures over years/months should be used to highlight potential problems and changes to system operation.

Title: Maintenance Log Books	Item: 7.1.3
Category: Introduction	Page: 1 of 1

In addition to the energy usage logs, a general log of preventative and remedial maintenance, should also be kept to assist with the diagnosis of potential problems and to provide a history of maintenance for future personnel. The exact arrangement of this maintenance log will be dependant on the preferences and regime implemented by the maintenance personnel, however, a logical approach to the record keeping is essential, particularly to cater for staff changes.

There are three main options with regards to the structure of record keeping as follows:-

- | | | |
|----|------------------|--|
| 1) | Chronological | All maintenance actions are recorded in chronological order. |
| 2) | System by System | Separate log books are kept for each system. |
| 3) | Equipment Basis | Separate log books are kept for each item of equipment. |

The advantages/disadvantages of each approach are summarised in the following table:-

Type	Advantages	Disadvantages
Chronological	Extremely simple and compact method of keeping records.	Locating historical information for any given item of equipment could prove difficult.
System	Historical comparison and identification of potential problems easily achieved.	Slightly more complex and bulky system of keeping records than the simple chronological approach.
Equipment	Greatest flexibility regarding historical comparison.	No more complex than the system approach, however, "documentation overload" could occur due to the sheer amount of paperwork being kept. Cross references between one item of equipment and another on the same system could prove difficult to maintain.

Title: Working Alone	Item: 7.2.1
Category: Risk Assessments	Page: 1 of 1

Description of Risk	Assessment of Risk		
	Low	Medium	High
Falling from a height		✓	
Electric shock		✓	

Actions to be Taken to Minimise the Risk	
Planning	Correct planning should include an assessment of the work, so that the likelihood of injury and the possible consequences to the lone worker are considered. Where there are hazardous substances, live electrical work, the need to work at height or near water, lone working should be avoided.
Physical	Lone worker alarms or other means of communication should be provided. First aid facilities to treat minor injuries should be available during work periods and those working alone should be advised of their location. Suitable means of access should be provided which can be handled safely by one person.
Managerial	Peripatetic workers should be supervised on a regular basis and management should ensure that their whereabouts are known at all times. Only experienced and trained operatives, with no adverse medical history, should be considered for working alone. The permitted limits of work and the limits on the initiative of the individual should be clearly specified before work is authorised.
Training	Levels of training and experience for lone workers should include a full understanding of the work involved, together with a thorough knowledge of the hazards and emergency procedures. Workers should also clearly understand the limits of the work which have been authorised and the limits on their own initiative.

Title: Working in Occupied Premises	Item: 7.2.2
Category: Risk Assessments	Page: 1 of 1

Description of Risk	Assessment of Risk		
	Low	Medium	High
Electric shock/burns		✓	
Fire		✓	
Falling from heights		✓	
Injuries to third parties - falls		✓	
Injuries to third parties - falling objects	✓		

Actions to be Taken to Minimise the Risk	
Planning	Planning should include the exchange of information with owner/occupier to ensure full reciprocal knowledge of existing hazards, demarcation of areas of responsibility and work hazards. Access equipment should be provided to ensure the maximum safety of workers and occupants. Details of existing services should be obtained before the commencement of works and COSHH and noise assessments should be available for materials and tools to be used. Physical barriers and notices should be installed to isolate work from occupants and members of the public. Fire exit routes should be kept free from obstruction and alternate routes should be clearly signed. Hot work should be controlled and extinguishers on hand. Where work at height needs to be carried out, debris netting, fans or other suitable measures to protect the public should be installed. Flammable and hazardous materials should be correctly controlled and stored.
Managerial	Supervisors should monitor all works, carrying out initial checks to ensure that safe systems of work are in place before commencement and that areas are left safe at the end of each work period. Regular liaison should take place with occupants to co-ordinate work and eliminate hazards to them. Use of personal protective equipment and ventilation equipment should be monitored to ensure compliance with COSHH assessments.
Training	Induction training should include any hazards and necessary precautions required for the workplace. Operative training should include safe systems of work and precautions designed to prevent injury to third parties. Supervisory management should be trained in site safety supervision.

Title: Working in Confined Spaces	Item: 7.2.3
Category: Risk Assessments	Page: 1 of 1

Description of Risk	Assessment of Risk		
	Low	Medium	High
Poisoning from toxic gases			✓
Asphyxiation - lack of oxygen			✓
Explosion			✓
Fire			✓
Excessive heat		✓	
Drowning			✓

Actions to be Taken to Minimise the Risk	
Planning	Planning should eliminate the need for entry or use of hazardous materials by the selection of alternative methods of work or materials. An assessment of the ventilation available and possible local exhaust ventilation requirements should be made, together with the potential presence of hazardous gases/atmosphere, process by products and the need to improve hygiene/welfare facility.
Physical	A documented entry system should be applied, preferably a Permit to Work System and adequate ventilation should be present. Detection equipment should be present before entry to check on levels of oxygen and presence of toxic or explosive substances. The area should be tested before entry and continually whilst any person is in the confined space. Breathing apparatus or airlines should be provided if local ventilation is not possible. Where no breathing apparatus is assessed as being required, emergency breathing apparatus and rescue harnesses should be provided. Rescue equipment, including lifting equipment, resuscitation facilities, safety lines and harnesses should be provided and a communication system with those in the confined space should be established. Air should not be sweetened with pure oxygen and precautions for the safe use of any plant or heavier-than-air gases in the confined space should be established before entry. Necessary personal protective equipment and hygiene facilities should be provided for those entering sewers.
Managerial	The management's role should be to decide on the nature of the confined space and put a safe system into operation, including checking the above. Flood potential and isolations should be checked.
Training	Full training should be provided for all persons entering and managing confined spaces. A surface rescue party should be trained in first aid procedures and the operation of testing equipment. All operatives and supervisory staff should be certified as trained to the same standard.

Title: Working at Heights	Item: 7.2.4
Category: Risk Assessments	Page: 1 of 1

Description of Risk	Assessment of Risk		
	Low	Medium	High
Falls of persons		✓	
Falls of materials	✓		

Actions to be Taken to Minimise the Risk	
Planning	Work should be planned to ensure that a safe means of access is provided. Reference should also be made to the use of access equipment. All equipment should be provided and maintained to the required legal standards..
Physical	Suitable signs and barriers should be positioned directly below the works to warn of overhead operations. Edge protection should be erected at places of work, where falls of more than 2 metres could occur. Where edge protection is removed for access, or is not reasonably practicable, personal suspension equipment should be used by operatives working at or near the edge. Where there is likely to be falling debris, chutes or full enclosures should be used to protect third parties. All operatives working below overhead operations should wear safety helmets.
Managerial	All equipment used should be checked to ensure it is in good order, to correct specification and in date for inspection. Work should be monitored to ensure that additional precautions and equipment is taken into use if edge protection is removed. Other relevant British Standards include: BS 7883:1997 - Anchors for Safety Harnesses, BS 8213 Part 1:1991 - Code of Practice for Cleaning of Windows and Doors, BS EN 355, 358, 361-365 - Personal Protective Equipment Against Falls from a Height.
Training	Training and instruction should be provided to all operatives and supervisory staff involved in the use of suspension equipment, such as lines and harnesses and how to inspect and assess personal protective equipment of this type before use.

Title: Use of Ladders	Item: 7.2.5
Category: Risk Assessments	Page: 1 of 1

Description of Risk	Assessment of Risk		
	Low	Medium	High
Falls of persons from ladders		✓	
Ladder slipping		✓	
Objects dropping by ladder user	✓		

Actions to be Taken to Minimise the Risk	
Planning	Ladders should be checked to ensure correct length, type and condition before use and should be subject to a planned maintenance programme. Ladder work should be restricted to that which can be carried out using one hand only and will only take twenty minutes to complete.
Physical	The ground base for the ladder should be firm and level. The ladder should be of sufficient length to extend a sufficient height above the step-off point when used as access to a scaffold. The correct angle of rest for a ladder is 75 degrees, or a base to height ratio of 1:4. Ladders should be secured against slipping, by tying at the top or the bottom. Ladders should only be footed as a sole precaution against movement if less than 3 metres high. Overreaching from ladders should be avoided. When using stepladders the hips should be kept below the top platform and between the stiles to ensure stability.
Managerial	Supervisors should check ladders before use to ensure they are sound. Use made of ladders should be monitored regularly, to ensure that operatives are not overreaching or using two hands to work. Damaged ladders should be broken up or removed from the workplace immediately. Painted ladders should not be accepted for use. Managers should check method statements supplied by sub-contractors and others, e.g. window cleaners, to ensure that ladders will be used correctly and that safe access will be available.
Training	All operatives should be trained in the safe use of ladders and the hazards which must be avoided. This should be done at induction.

Title: Use of Trestles	Item: 7.2.6
Category: Risk Assessments	Page: 1 of 1

Description of Risk	Assessment of Risk		
	Low	Medium	High
Falls of persons and materials from working platform		✓	
Collapse of trestles		✓	

Actions to be Taken to Minimise the Risk	
Planning	Lightweight staging or mobile access scaffold should be used in preference to trestles. Trestles should only be planned for light work of relatively short duration.
Physical	Lightweight staging should be used in preference to boards and be positioned so that at least one-third of the height of the trestle is above the platform. Staging/platform width should be at least 600mm. Toe-boards and guardrails are required to platforms above 2 metres in height.
Managerial	All equipment used should be checked to ensure it is in good order before use, to ensure no loose or damaged cross bearers, the hinges are in good order and that there are no defective stiles (loose or missing bolts or screws). Management should supervise the use of trestles to ensure that operatives moving and erecting trestles are properly trained and instructed and that those using trestles do not overreach.
Training	Training and instruction should be provided to all operatives and supervisory staff involved in the use of trestles as above.

Title: Use of Hand Tools	Item: 7.2.7
Category: Risk Assessments	Page: 1 of 1

Description of Risk	Assessment of Risk		
	Low	Medium	High
Eye Injury		✓	
Injury to hands, feet and body	✓		

Actions to be Taken to Minimise the Risk	
Planning	Tools provided by the employer should be assessed to ensure they are fit for the environment in which they are to be used and in good working condition. Tools are required to be suitable for the purpose for which they will be used.
Physical	Eye protection should be provided and used whenever work is carried out using cold chisels, drills, grinders or other tools where there is a risk of flying particles or pieces of the tool breaking off. Open-bladed knives, screwdrivers and other sharp tools should be carried and used so as not to cause injury to the user or others. Non-ferrous (spark-free) tools should be used in flammable atmospheres. Insulated tools should be used where there is a possibility of live electrical work.
Managerial	Management should monitor hand tools which can deteriorate with use, to ensure they are sharpened or replaced as necessary and that the correct tools are being used properly. Specific checks should be made as follows: Chisels for mushroom heads Hammer and file handles for deterioration and exposed tangs Open-ended spanners for splayed jaws.
Training	Operative should be instructed in the correct method of use and in maintenance requirements at induction if not part of craft training.

Title: Introduction	Item: 7.3.1
Category: C.O.S.H.H. Instructions	Page: 1 of 1

The Control of Substances Hazardous to Health (C.O.S.H.H.) 2002 regulation place a duty on employers to ensure that their employees are not exposed to substances hazardous to health, ie. to solids/liquids/gases that are toxic/harmful/corrosive/irritant. Safety guidelines, for example, for concentrations of substances in air, are separately prescribed by the appropriate authorities.

The following pages within this sub-section give a general indication of the potential hazards presented and procedures for the safe storage, handling and disposal of the various chemicals and materials likely to be encountered during the operation and maintenance of the system.

It is essential however that the specific C.O.S.H.H. data sheet (also known in some instances as a Chemical Safety Data Sheet (C.S.D.S) or a Material Safety Data Sheet or M.S.D.S.) for any chemicals or materials used are acquired, read, understood and followed by all personnel. The purpose therefore of this sub-section is for planning information to ensure that all necessary procedures are in place prior to the purchase of the chemicals or materials.

Title: Refrigerant Gas	Item: 7.3.2
Category: C.O.S.H.H. Instructions	Page: 1 of 1

Description: This describes the general procedures for the safe storage, handling and disposal of refrigerant gasses, together with how to treat contamination.

Health Hazard Data:

Skin Contact: Liquid splashes or sprays may cause freeze burns.

Eye Contact: Liquid splashes or sprays may cause freeze burns.

Inhalation: Low toxicity with normal concentrations, high exposures can cause abnormal heart rhythm and prove suddenly fatal. Very high exposures may cause anaesthetic effects and asphyxiation.

Notes:

1. It is essential that the specific C.O.S.H.H. data sheet for the gas to be used is acquired, read, understood and followed by all personnel prior to its use.
2. If there is any doubt as to the subjects condition then medical attention should be sought immediately.

Procedure	Actions	Notes
Storage	The canisters should be stored in a cool (below 45°C), dry, well ventilated space away from sunlight and sources of heat. The canisters should be stored away from ventilation intakes of any system and away from open drains.	
Handling	Use specialised refrigerant charging equipment and suitable respiratory equipment in unventilated spaces. Charging should be made using the liquid phase and not the vapour phase.	Refrigerant transfers can cause a build up of static electricity, all apparatus should be earthed. In the event of leakage it should be noted that refrigerant gasses will accumulate at floor level.
Disposal	Unwanted refrigerant gas should be collected for recycling if at all possible, or else destroyed by a suitably licensed agent.	
Skin Contact	Thaw affected areas with warm water, remove contaminated clothing.	Clothing may be frozen to skin, thaw first.
Eye Contact	Remove any contact lenses and wash eyes with copious amounts of fresh water.	Behind the eye lids should also be washed for 15 minutes. Seek medical attention immediately.
Inhalation	Relocate the subject to the fresh air and perform artificial respiration if necessary.	Seek medical attention immediately.
Ingestion	Do not induce vomiting. If conscious drink 250ml of water and keep warm.	Seek medical attention immediately.

Title: Foam Insulation	Item: 7.3.3
Category: C.O.S.H.H. Instructions	Page: 1 of 1

Description: This describes the general procedures for the safe storage, handling and disposal of foam based insulating materials, together with how to treat contamination.

Health Hazard Data:

Skin Contact: No observed hazards.

Eye Contact: In particulate form irritation may occur.

Inhalation: Potential irreversible effects if burnt.

Notes:

1. It is essential that the specific C.O.S.H.H. data sheet for the material to be used is acquired, read, understood and followed by all personnel prior to its use.
2. If there is any doubt as to the subjects condition then medical attention should be sought immediately.

Procedure	Actions	Notes
Storage	There are no special storage requirements for this product.	
Handling	Gloves and goggles should be used when installing.	
Disposal	General good housekeeping measures should prove sufficient. Controlled combustion and burial are required for large quantities.	
Skin Contact	Any contact with burning material should be doused with cold water, but not removed.	Seek medical attention immediately.
Eye Contact	Treat as any particulate such as dust or grit.	
Inhalation	Not applicable unless burnt.	Seek medical attention immediately.
Ingestion	Do not induce vomiting.	Seek medical attention immediately.

SECTION SEVEN

Title: General Maintenance Log Book	Item: 7.4.1
Category: Sample Log Books	Page: 1 of 1

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[illegible]

SECTION EIGHT

FAULT TRACIİ G CHARTS

INDEX

<u>ITEM</u>	<u>DESCRIPTION</u>
8.1	- <u>GENERAL</u>
8.1.1	- Introduction
8.1.2	- Preliminary Investigation
8.2	- <u>MISCELLANEOUS EQUIPMENT</u>
8.2.1	- Direct Expansion Air Conditioning Unit
8.2.2	- Compressor
8.2.3	- Fan
8.2.4	- Small Fan
8.2.5	- Coil (DX)
8.2.6	- Motor

SECTION EIGHT

Fault Tracing Charts

Title: Introduction	Item: 8.1.1
Category: General	Page: 1 of 1

Fault tracing of the electrical installation or appliances that are connected to the installation should only be carried out by suitably trained competent personnel.

Whenever possible all fault tracing should be carried out on circuits, appliances or components after they have been suitably isolated from the main supply.

When it is not possible to carry out fault tracing on a "dead" installation, then the fault tracing must be carried out by a minimum of two people.

Prior to carrying out any fault tracing it is essential that the personnel involved read the health and safety section of this manual and familiarise themselves with the operation and maintenance sections.

Title: Preliminary Investigation	Item: 8.1.2
Category: General	Page: 1 of 1

Following a reported failure of an appliance, the likely cause can be quickly established by asking the relevant question of those in the vicinity of the failure.

- a) Was the appliance or component operating when it failed, or did it fail to start?
- b) Do other items of equipment on the same circuit still operate?
- c) Has the correct operating procedure been carried out?
- d) When was the last time the component/appliance was used?
- e) Had there been any signs of deterioration in the performance or any increase in noise levels from the appliance/component?
- f) Has there been any maintenance work carried out prior to the failure?
- g) Has any work been carried out in the vicinity of the failure, or along the route of the failed circuit prior to the failure?
- h) Has anyone else investigated the failure prior to those now required to do so and if so what did they do and what did they find out?

SECTION EIGHT**Fault Tracing Charts**

Title: Direct Expansion Air Conditioning Unit - Indoor Unit	Item: 8.2.1
Category: Miscellaneous Equipment	Page: 1 of 4

Symptom	Possible Cause	Checks
Fan Not Working	Electrical	Check that electrical supply is available at fan.
		Check local isolator is switched on.
		Check for loose electrical connections.
		Check controller is calling for fan to run.
	Drive Mechanism	Check motor is working.
		Check if fan blowers are aligned.
		Check inlet and outlet of fan for obstruction.
Duty Low	Electrical	Check for correct supply voltage.
		Check for correct control set points.
	Resistance	Check condition of air filter.
		Check condition of evaporator fins for fouling.
		Check for correct system balance.
	Refrigerant	Check for leaks in systems.
		Check for excessive high head pressure.
		Check for blocked expansion valve orifice.
Duty High	Electrical	Check for correct supply voltage.
		Check for correct controls set points.
Excessive Vibration	Resistance	Check correct system balance.
	Drive Mechanism	Check for correct balance of fan impeller.
		Check condition of bearings.
		Check for drive shaft distortion.
		Check motor balance.
	General	Check security of casing.
		Check security of pipework.

SECTION EIGHT**Fault Tracing Charts**

Title: Direct Expansion Air Conditioning Unit - Indoor Unit	Item: 8.2.1
Category: Miscellaneous Equipment	Page: 2 of 4

Symptom	Possible Cause	Checks
Excessive Vibration (Continued)	Mountings	Check security of mounting bolts.
		Check condition of anti-vibration mounts.
Excessive Noise	Drive Mechanism	Check condition of bearings.
	Damage	Check condition of motor.
		Check impeller for impingement on casing.
	Vibration	Refer to previous symptom.
Water Leakage	Blocked Drain	Check condensate tray for obstruction.
		Check condensate drain for obstruction.

SECTION EIGHT**Fault Tracing Charts**

Title: Direct Expansion Air Conditioning Unit - Outdoor Unit	Item: 8.2.1
Category: Miscellaneous Equipment	Page: 3 of 4

Symptom	Possible Cause	Checks
Fan Not Operating	Electrical	Check that electrical supply is available at fan.
		Check that local isolator is switched on.
		Check for loose electrical connections.
		Check controller is calling for fan to run.
Compressor Not Operating	Drive Mechanism	Check motor is working.
		Check fan is free to rotate.
		Check that electrical supply is available at compressor.
		Check isolator is on.
Excessive Vibration	Electrical	Check for loose electrical connections.
		Check for compressor motor failure.
	Mechanical	Check for compressor seizure.
		Check for correct balance of fan impeller.
Excessive Noise	Drive Mechanism (Fan)	Check condition of bearings.
		Check condition of motor.
	Compressor Fault	Check oil level.
		Check for broken or damaged compressor valve reeds.
		Check for worn or scored bearings.

SECTION EIGHT**Fault Tracing Charts**

Title: Direct Expansion Air Conditioning Unit - Outdoor Unit	Item: 8.2.1
Category: Miscellaneous Equipment	Page: 4 of 4

Symptom	Possible Cause	Checks
Excessive Noise Continued	Damage	Check fan impeller for impingement on casing.
	Vibration	Refer to previous symptom.
Excessive Head Pressure	Fan Not Working	Refer to previous symptom.
	Refrigerant	Check for overcharge of gas.
		Check for contaminants in refrigerant lines.
	General	Check condenser coil for obstruction.
Low Suction Pressure	Refrigerant	Check piping for leaks.
		Check refrigerant charge.
		Check for obstruction in flow control device.
		Check for clogged filter drier.
	General	Check for obstruction or icing on coil.
Excessive Suction Pressure	Refrigerant	Check refrigerant charge.
		Check flow through flow control device.
	Controls	Check setting of head pressure controller.
	Compressor	Check for damage to compressor reeds.

SECTION EIGHT**Fault Tracing Charts**

Title: Compressor	Item: 8.2.2
Category: Miscellaneous Equipment	Page: 1 of 1

Symptom	Possible Cause	Checks
Will Not Operate	Electrical	Check if electrical supply is available at unit.
		Check if local isolator is switched on.
		Check for correct electrical supply voltage.
	Controls	Check controls are calling for compressor to run. Check for correct operation of all control components.
Stops	Motor	Check for a seized or burnt out motor.
	Mechanical	Check for locked or damaged components.
	Electrical	Check for a loss of electrical supply.
	Controls	Check for safety cut-out in controls.
Short Cycles	Pipeline	Check refrigerant charge.
	Compressor	Check oil level.
	Controls	Check for correct operation of controls.
	Motor	Motor faulty.
Runs Continuously	Pipeline	Check condition of drier core.
		Check condition of solenoid valves.
		Check refrigerant charge.
	Controls	Check for correct operation of controls.
Noisy	Compressor	Check oil level.
		Check for broken internal components.
	Pipeline	Check for correct operation of solenoid valves.
	Controls	Check for correct operation of control items.

SECTION EIGHT**Fault Tracing Charts**

Title: Fan	Item: 8.2.3
Category: Miscellaneous Equipment	Page: 1 of 1

Symptom	Possible Cause	Checks
Not Running	Electrical Supply	Check that electrical supply is available at fan. Check local isolator is switched on. Check controller is calling for fan to run.
	Drive Mechanism	Check motor is working.
Duty Low	Electrical Supply	Check for correct supply voltage. Check for correct speed setting.
	Resistance	Check inlet and outlet of fan for obstruction. Check for correct system balance.
Duty High	Electrical Supply	Check for correct supply voltage. Check for correct speed setting.
	Resistance	Check for correct system balance.
Excessive Vibration	Drive Mechanism	Check for correct balance of fan impeller. Check condition of bearings. Check for drive shaft distortion. Check motor balance.
	Mountings	Check security of holding down bolts. Check condition of anti-vibration mounts.
Excessive Noise	Drive Mechanism	Check condition of bearings. Check condition of motor.
	Damage	Check impeller for impingement on casing.
	Vibration	Refer to previous symptom.

SECTION EIGHT**Fault Tracing Charts**

Title: Small Fan	Item: 8.2.4
Category: Miscellaneous Equipment	Page: 1 of 1

Symptom	Possible Cause	Checks
Not Running	Electrical Supply	Check that electrical supply is available at fan.
		Check local isolator is switched on.
		Check controller is calling for fan to run.
	Drive Mechanism	Check motor is working.
Duty Low	Electrical Supply	Check for correct supply voltage.
		Check for correct speed setting.
	Drive Mechanism	Check impeller is secure on shaft.
	Resistance	Check inlet and outlet of fan for obstruction.
Duty High	Electrical Supply	Check for correct supply voltage.
		Check for correct speed setting.
Excessive Noise	Drive Mechanism	Check impeller is secure on shaft.
		Check condition of motor.
	Damage	Check impeller for impingement on casing.

SECTION EIGHT**Fault Tracing Charts**

Title: Coil (DX)	Item: 8.2.5
Category: Miscellaneous Equipment	Page: 1 of 1

Symptom	Possible Cause	Checks
Excessive Cooling	Controls	Check for correct operation of thermostats and sensors. Check for correct response at controller.
	Refrigerant	Check for correct operating pressure.
Insufficient Cooling	Controls	Check for correct operation of thermostats and sensors. Check for correct response at controller.
	Refrigerant	Check for leaks in systems. Check for correct operating pressure.

SECTION EIGHT

Fault Tracing Charts

Title: Motor	Item: 8.2.6
Category: Miscellaneous Equipment	Page: 1 of 6

The most common cause of faults that occur is due to motors of pumps or fans. This chart has therefore been structured differently from the rest so as to lead not only to the cause but also to the remedy.

Step	Type	Details	Answer	Go To Step
1	Test	Are all isolators switched on.	Yes	4
			No	2
2	Probable Fault	Switch isolators on.	-	3
3	Test	Does motor start to run.	Yes	END
			No	4
4	Test	Are any fuses ruptured.	Yes	5
			No	8
5	Test	Are there any short circuits between phase to neutral or phase to earth.	Yes	7
			No	6
6	Action	Replace ruptured fuses.	-	3
7	Probable Fault	Repair short circuits.	-	6
8	Test	Are all phase voltages satisfactory.	Yes	13
			No	9
9	Test	Check all phase voltages back to incoming supply.	-	10
10	Test	Are all phase voltages satisfactory at incoming supply.	Yes	12
			No	11
11	Action	Refer to local electricity board.	-	END
12	Probable Fault	Correct unsatisfactory voltage checking for loose connections, etc.	-	8
13	Test	Is starter tripped on a thermal overload condition.	Yes	14 & 15
			No	23
14	Action	Reset the overload current.	-	3

SECTION EIGHT**Fault Tracing Charts**

Title: Motor	Item: 8.2.6
Category: Miscellaneous Equipment	Page: 2 of 6

Step	Type	Details	Answer	Go To Step
15	Action	Measure the output current and voltage from starter.	-	16
16	Test	Are the voltages equal.	Yes	17
			No	22
17	Test	Are overloads set correctly.	Yes	18
			No	21
18	Test	Is the current consumed:-	-	-
		1) Fractionally higher than motor rating.	Yes	19
		2) Excessively higher than motor rating.	Yes	20
19	Action	Set overload unit slightly higher and allow motor to run for 15 minutes.	Yes	END
		Does motor run satisfactory without overheating.	No	20
20	Probable Fault	Check that motor is not excessively overloaded. If so reduce the loading so the motor is operating to its designed conditions.	-	13
21	Action	Adjust overload unit to correct setting.	-	13
22	Action	Trace exactly where the voltage drop occurs and remedy by cleaning contacts.	-	13
23	Test	Is the starter equipped with a thermistor protector.	Yes	24
			No	28
24	Test	Is the thermistor protection tripped.	Yes	25
			No	28
25	Test	Is the wiring of the thermistor and the resistance of the thermistor circuit satisfactory.	Yes	27
			No	26
26	Probable Fault	Motor mounted thermistor may be burnt out or wiring may be broken.	-	END

SECTION EIGHT**Fault Tracing Charts**

Title: Motor	Item: 8.2.6
Category: Miscellaneous Equipment	Page: 3 of 6

Step	Type	Details	Answer	Go To Step
27	Action	Reset thermistor protector unit.	-	3
28	Test	Is the starter equipped with an earth leakage protector circuit.	Yes	29
			No	33
29	Test	Is the earth leakage protector tripped.	Yes	30
			No	33
30	Test	Is there an earth leakage fault or any wiring fault on the earth leakage circuit.	Yes	31
			No	32
31	Probable Fault	Earth Leakage.	-	END
32	Action	Reset earth leakage protector unit.	-	3
33	Test	What type of starter is used.	D.O.L.	34
			A.S.D.	37
			Stator Rotor	41
			Auto-Trans	44
			Vapour-matic	34
34	Test	Does the starter contactor operate.	Yes	47
			No	35
35	Test	Is there any voltage across the contactor coil terminals.	Yes	36
			No	48
36	Probable Fault	Contactor coil burnt out.	-	END

SECTION EIGHT**Fault Tracing Charts**

Title: Motor	Item: 8.2.6
Category: Miscellaneous Equipment	Page: 4 of 6

Step	Type	Details	Answer	Go To Step
37	Test	Do the starter contactors operate:- Main Contactor	Yes	39
			No	35
		Star Contactor	Yes	39
			No	38
		Delta Contactor	Yes	39
			No	38
38	Test	Is there any voltage across the contactor coil terminals.	Yes	36
			No	39
39	Test	Check timer unit for correct action remembering when initially started the Star and Main Contactors should be energised and at the end of the time period the Delta and Main Contactors should be energised.	Correct	49
			Incorrect	40
40	Probable Fault	Timer unit may be faulty or wiring associated may be faulty	-	END
41	Test	Do the starter contactors operate:- Main Contactor	Yes	43
			No	35
		Subsidiary Contactors	Yes	43
			No	42
42	Test	Is there any voltage across the contactor coil.	Yes	36
			No	43
43	Test	Check timing circuits for correct action remembering when initially started the Main Contactor should be energised and at the end of each time period a subsidiary contactor will also be energised, until all are energised.	Correct	49
			Incorrect	40

SECTION EIGHT**Fault Tracing Charts**

Title: Motor	Item: 8.2.6
Category: Miscellaneous Equipment	Page: 5 of 6

Step	Type	Details	Answer	Go To Step
44	Test	Do the starter contactors operate:-	Yes	46
		Main Contactor	No	35
		Subsidiary Contactors	Yes	46
			No	45
45	Test	Is there any voltage across the contactor coil terminals.	Yes	36
			No	46
46	Test	Check timing circuits for correct action remembering when initially started the Star and Load Contactors should be energised and at the end of the time period the Run Contactor should be energised whilst the Star and Load Contactors should be de-energised.	Correct	49
			Incorrect	40
47	Test	Check the control action of the starter.	Correct	49
			Incorrect	48
48	Probable Fault	Control circuit wiring of the started may be faulty.	-	END
49	Test	Are the motor windings open circuit.	Yes	53
			No	50
50	Test	Are the motor windings of equal resistance.	Yes	51
			No	53
51	Test	Is there continuity through all the contactors when contacts are closed.	Yes	52
			No	53
52	Test	Is the wiring between the starter and the motor satisfactory, i.e. are there any breakages or high resistances.	Satisfactory	56
			Not Satisfactory	55
53	Probable Fault	Any one of the motor windings being open circuit or of different resistance would result in an unbalance load.	-	END

SECTION EIGHT**Fault Tracing Charts**

Title: Motor	Item: 8.2.6
Category: Miscellaneous Equipment	Page: 6 of 6

Step	Type	Details	Answer	Go To Step
54	Probable Fault	Discontinuity through contacts would not allow the motor to run.	-	END
55	Probable Fault	Broken wiring or high resistance wiring would result in an unbalance load.	-	END
56	-	Problem not found - Go back to Step 1.	-	1

SECTION II E

MANUFACTURERS' LITERATURE

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SECTION TWELVE

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12.1.1	-	Schedule of Record Drawings

SECTION TWELVE

Record Drawings

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Category: Record Drawings	Page: 1 of 1

Number	Size	Title
RD01	A1	Ground Floor Reception Mechanical Services Layout
RD02	A1	Roof Level Mechanical Services Layout